

MUHAMMAD AZIZ

JUNIOR MACHINE LEARNING ENGINEER

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Informatics Engineering student and software developer with hands-on experience in full-stack web development, AI/ML, computer vision, IoT, mobile applications, and system administration. Experienced in developing academic, freelance, organizational, and competition-based projects using React, Laravel, TypeScript, Python, FastAPI, Flutter, Firebase, and machine learning technologies. Interested in building practical digital solutions and leading technology projects from development to deployment.

WORK EXPERIENCE

Web Developer & Digital Hub Division Komit PNL May 2026 – Present

- Develop and maintain web applications and digital platforms for organizational and academic needs.
- Contribute to the development of Komit PNL digital initiatives and web platforms.
- Deliver web development activities using modern frontend and backend technologies.
- Serve as a web development mentor and speaker for programming activities within Komit PNL.

Freelance Web Developer 2024 – Present

- Develop responsive websites and web applications for personal, academic, and client projects.
- Build landing pages, management systems, e-commerce platforms, and information systems using modern web technologies.
- Deliver projects based on client requirements, including UI implementation, backend integration, database design, and deployment.

Web Developer & Technical Mentor | Freelance / Komit PNL July 2026

- Mentor students and peers in web development fundamentals and practical project development.
- Deliver technical sessions covering frontend development, backend integration, Git, databases, and deployment.
- Guide project development from UI implementation to functional web applications.

SELECTED FREELANCE PROJECTS

PFASmartAI - AI Learning Reflection Platform August 2026 - Present

- Developed an AI-assisted student reflection platform connecting student check-ins with teacher analytics.
- Implemented student reflection flows, AI conversation, emotional-zone classification, and teacher dashboards.
- Integrated real-time analytics, reporting, and student monitoring workflows.

CocoCarbone - Company Profile & Export Landing Page May 2026 - July 2026

- Developed a responsive company profile and export-focused landing page for CV Gemilang Indo Nusantara.
- Implemented bilingual presentation, responsive/mobile-first design, product presentation, and WhatsApp inquiry integration.
- Built the interface to support international B2B audiences.

POS Application

December 2025 - April 2026

- Developed a real-time Point of Sales system with mobile cashier and web dashboard.
- Implemented product management, transactions, authentication, and automated reporting.

SELECTED TECHNICAL PROJECTS

PETROCHAIN - AI / Computer Vision / Blockchain / IoT

- Developed a prototype for targeted and auditable subsidized-fuel distribution.
- Implemented vehicle classification, YOLO-based detection, OCR validation, eligibility filtering, and blockchain audit trails.
- Technologies: YOLO, OCR, XGBoost, Blockchain, ESP32, IoT

ByteShield - Static Malware Detection - Deep Learning / Cybersecurity

- Developed a static malware-analysis prototype for Windows executable files without executing the binaries.
- Built a machine-learning pipeline for executable analysis, malware classification, and result visualization.
- Technologies: Python, CNN, FastAPI, React, Deep Learning

SDG Sentiment Analysis - Machine Learning / NLP

- Built a sentiment-analysis dashboard for public discussions related to the Sustainable Development Goals.
- Analyzed sustainability-related conversations and presented insights through interactive visualizations.
- Technologies: Python, NLP, Pandas, Plotly, Streamlit

ACHIEVEMENTS

1st Place - E-Government, KMIPN VIII 2026, Politeknik Negeri Lhokseumawe, 2026

Team Leader - PETROCHAIN, Intelligent Verification & Audit for Subsidized Fuel Distribution

Finalist - E-Government, KMIPN VIII 2026, Politeknik Negeri Lhokseumawe, 2026

TeamBerapa - PETROCHAIN

1st Place - Machine Learning & Data Science, Hackathon Bagus #3, Kelas Bagus, 2025

Team Member - SDG Sentiment Analysis

2nd Place - VibeCoding, FTP Policy PNL, Politeknik Negeri Lhokseumawe, 2026

ByteShield - Static Malware Analysis System

CERTIFICATIONS

- BNSP - Web Developer
- Dicoding - Belajar Dasar Data Science
- Dicoding - Belajar Dasar Cloud dan Gen AI di AWS
- Dicoding - Belajar Dasar Manajemen Proyek
- Kelas Bagus - Machine Learning / Hackathon Program

TECHNICAL SKILLS

Programming & Frameworks & Database

Laravel · PHP · Python · FastAPI · Firebase · MySQL · PostgreSQL · SQLite · REST API

AI / Machine Learning

Machine Learning · Deep Learning · CNN · Computer Vision · YOLO · OCR · NLP · Sentiment Analysis · XGBoost

DevOps & Infrastructure & IOT

Git · GitHub · Docker · VPS · Linux · Ubuntu Server · Windows Server Network Administration · Cryptography Fundamentals ESP32 · Arduino · Firebase · Sensors · Real-Time Monitoring

ORGANIZATIONAL EXPERIENCE

Komit PNL - Digihub / Web Development

May 2026 - Present

- Web development and digital initiatives.
- Technical mentoring and project development.
- Support technology-focused community activities.